Report: Iris Mechanism

A Mech Machinery (2103322) Project

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1. Introduction

An **Iris Mechanism** (sometimes known as an Aperture) is a mechanism that converts rotational motion into synchronized radial movement to open and close a central circular aperture.

This type of mechanism is mostly used in imaging technologies as an element that control how much light can go through the center hole. Due to its symmetrical nature, it can be used alongside glass elements to create an image with the desired exposure and/or depth of field.

2. Our Setup

This specific iris mechanism is hexagonal in shape. The reason it is not circular is due to ease of design and time constraint, as designing circular opening would be more difficult. Every translating joint in this design can be design linearly, improving design time.

The diameter of the central aperture is 50 mm, thus the diameter for a fully open aperture. The spin ring can rotate 35° counterclockwise, creating a trigonometry function that converts ring rotation into distance travel along each blade.

From the formula for degree(s) of freedom:

$$DoF = 3(L - 1) - 2J_1 - J_2,$$

with 8 links (Main structure/ground, Rotating disk, and 6 blades), 6 full joints (6 slotted holes), and 8 half joints (Sliding between the layers of disk and blades), we can confirm that this mechanism has 1 degree of freedom.

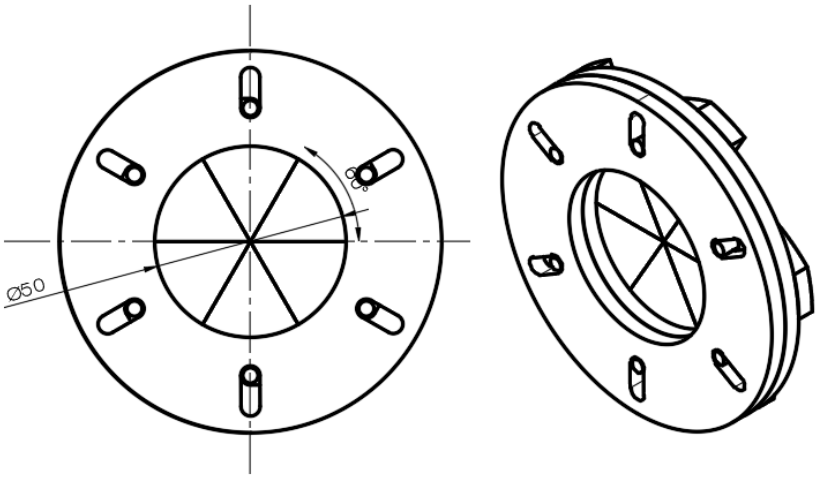


Figure 1: Iris Mechanism, fully closed.

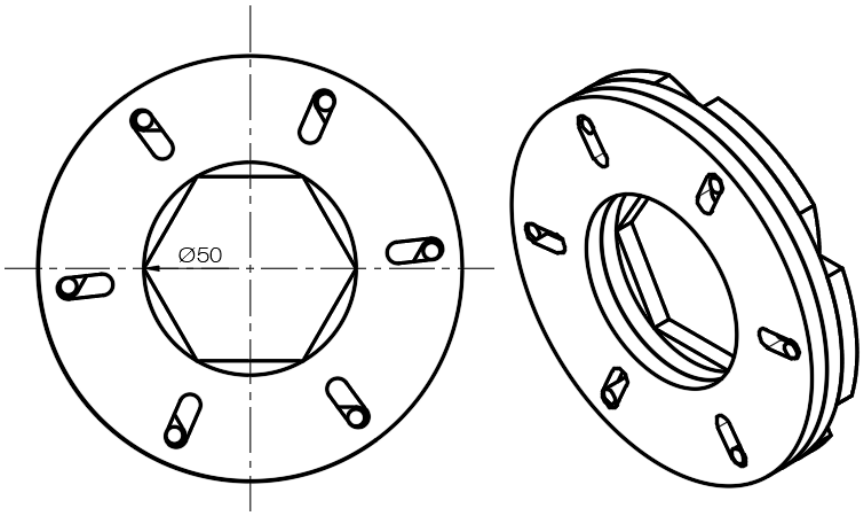


Figure 2: Iris Mechanism, fully opened.

3. Simscape Multibody

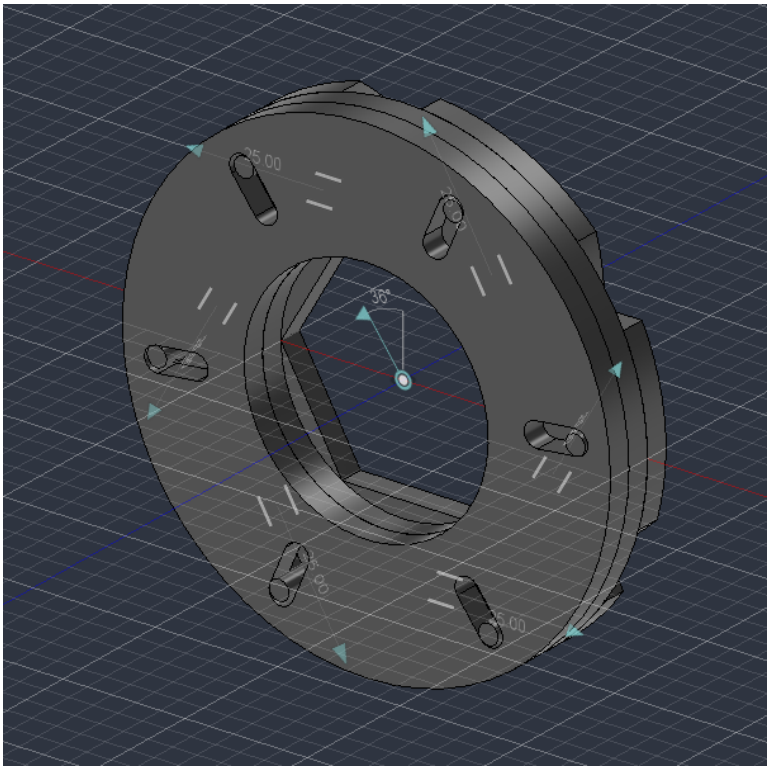


Figure 3: Fusion 360 model.

Start by building a simple model in fusion 360. Then, importing all the components into Simscape Multibody. Study the angle of rotation and how it opens the mechanism.

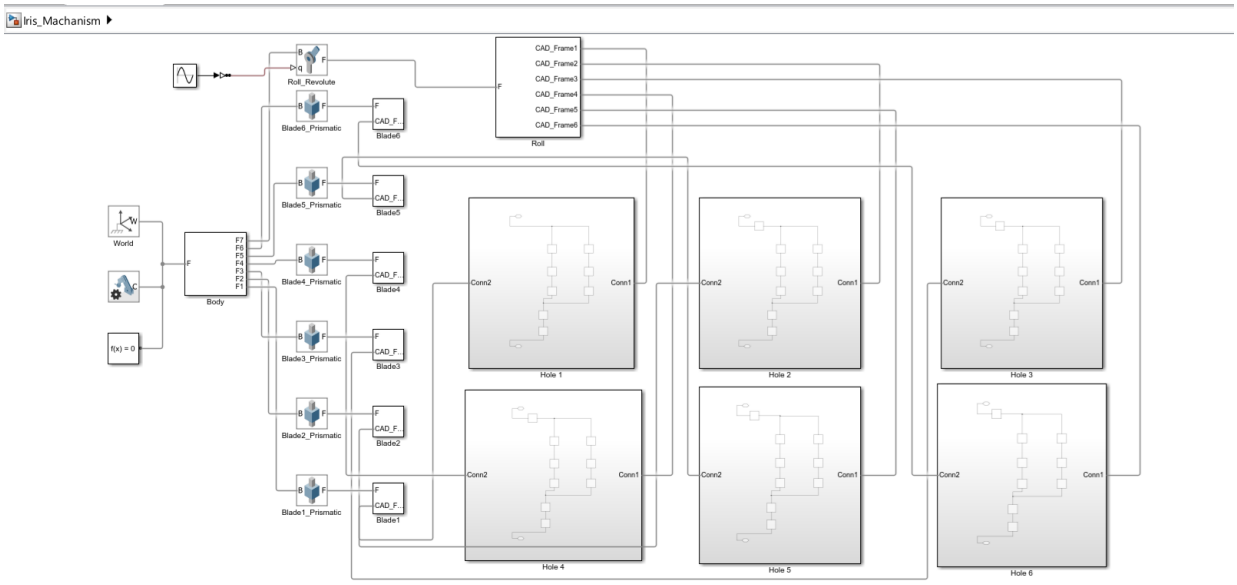


Figure 4: Entirety of the Simulink file

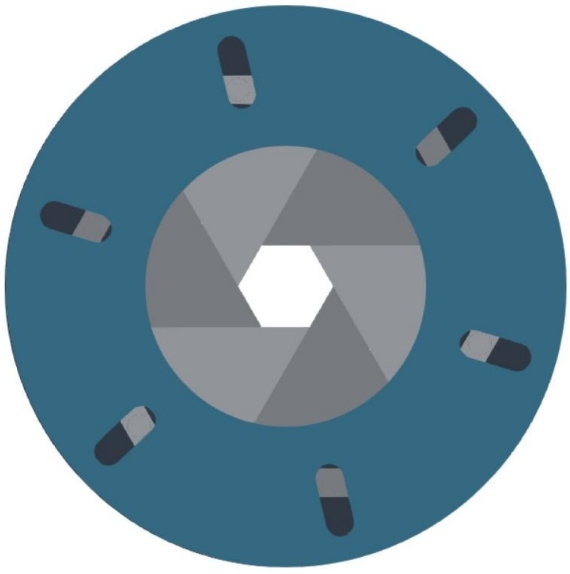


Figure 5: Simscape Multibody model in Explorer.

4. Symbolic Math

```
syms theta;
%%Plot how far the element moved
L = sin(theta)*87.17;
fplot(L,[0 35*2*pi/360]);
xlabel('\theta (rad)');
ylabel('Focal length at f/1 (mm)');
title('Focal length per unit rotation');

%%Plot Area
Area = 6*sqrt(3)/4 * (L/2)^2;
fplot(Area, [0 35*2*pi/360]);
xlabel('\theta (rad)');
ylabel('Area / \theta (mm^2)');
title('Area per unit rotation');

%%Plot f/ number
f = (6*sqrt(3)/4 * (25)^2)/Area;
fplot(f, [6*2*pi/360 35*2*pi/360]); %Skip first 6 degree as f/ reach inf (Area = 0)
xlabel('\theta (rad)');
ylabel('f/');
title('f/ number at f = 50mm');
```

Figure 6: MATLAB code

The MATLAB code used to analyze the mechanism consists of three parts.

4.1) Measure how far the mechanism moved

As the relation between the angle of rotation of the spin ring and the distance travel is a sin function of angle of rotation:

$$Focal\ Length = 50\sin(\theta)\ /\sin\ (35°)$$

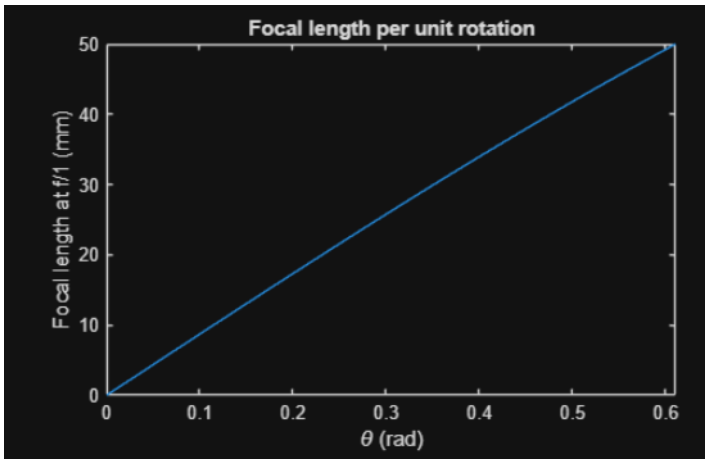


Figure 7: Focal length per unit rotation at f/1.

4.2) Calculate the area of the opened aperture

There are of the opened aperture can be calculated from the triangle formula multiplied by 6 for the 6 blades.

$$Area = 6 \times \frac{\sqrt{3}}{4} \times L^2$$

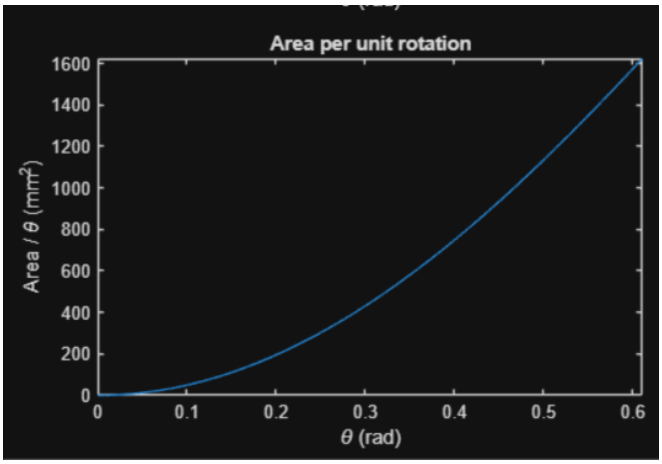


Figure 8: Area per unit rotation.

4.3) Finding f/ number

In the camera world the f/ number represents how much the aperture opens in relation to the focal length. It is represented as f/ followed by a number, meaning focal length divided by a number. This gives us the diameter of the hole. As the f/ number increases the aperture size would decrease.

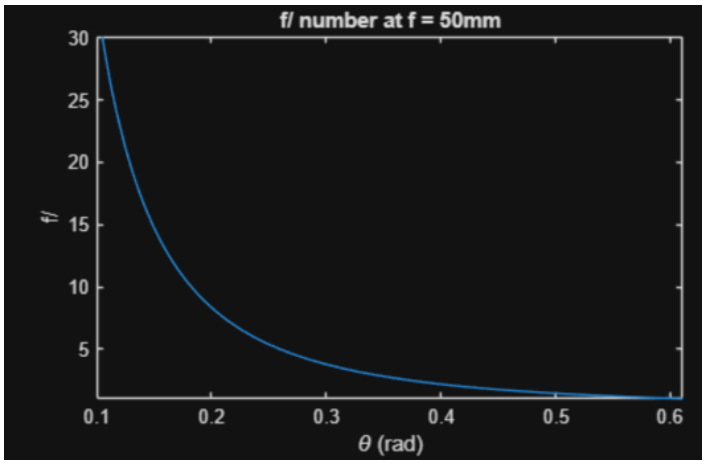


Figure 9: f/ number at f = 50mm for this iris mechanism.

5. Use in real life

In reality if the iris mechanism is designed as it is in our setup, the amount of half joint would cause a lot of jamming. So optical designers opted to making the blades of the iris mechanism literal blades, as they are very thin, making its operation as smooth as possible. The blades are also in the shape of a crescent and rotates slightly when operate rather than just sliding, creating a shape that is closer to a circle.

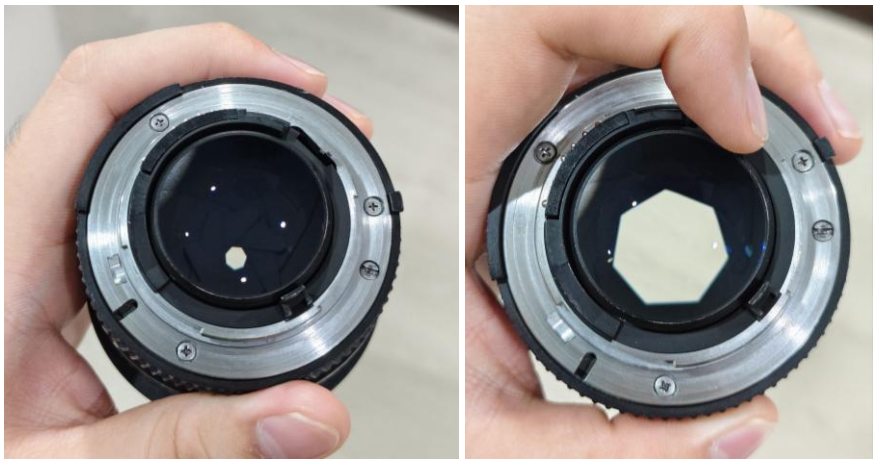


Figure 10: A real aperture mechanism in a Nikon 50mm f/1.4D lens